



# RAYCATCHER

## Smart Solar Tracking System



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### KEY PARTNERS



Solar EPC companies & installation contractors



IoT hardware suppliers (sensors, microcontrollers)



Cloud providers (AWS)



Govt schemes (PM Surya Ghar — 10M households targeted)



Solar panel manufacturers



Distribution partners (Flipkart, Amazon, local dealers)



### KEY ACTIVITIES

- Designing & manufacturing tracking hardware
- Developing IoT + cloud analytics platform
- Installation & retrofitting on existing solar panels
- Data monitoring & performance optimisation
- Customer support and maintenance
- Partner onboarding (EPC + distributors)



### VALUE PROPOSITION

- Increase solar energy generation by ~30% vs fixed-tilt (industry benchmark: 20–30%)
- Affordable alternative — commercial dual-axis trackers cost ₹5–10L+ ; RayCatcher targets ₹20–25K
- Retrofit solution (no need to replace panels)
- Real-time monitoring via IoT dashboard
- Improves ROI & reduces payback period — Indian solar payback currently ~3 years for C&I
- Smart automation (no manual tilt adjustments)
- Serves 20.8 GW cumulative rooftop base (India, 2025)



### KEY RESOURCES

- IoT hardware (sensors, motors, MCU)
- Cloud infrastructure (AWS — ₹5K/month)
- Engineering & product team
- Proprietary tracking algorithm (IP asset)
- Manufacturing setup — BOM ₹12–16K/unit
- Distribution network



### CUSTOMER RELATIONSHIPS

- Self-service dashboard (web + mobile)
- Automated alerts for faults & performance
- Subscription-based cloud analytics — target <10% annual churn
- Customer support & maintenance services
- Feedback-driven product improvements
- Target NPS >50 post-install



### CHANNELS

- Direct-to-consumer (website + ROI calculator)
- Solar EPC partnerships (B2B) — top 10 installers hold 26% of India market
- E-commerce platforms (Flipkart, Amazon)
- YouTube demos & digital marketing
- LinkedIn outreach for commercial buyers
- Govt program integrations (PM Surya Ghar)
- WhatsApp referral programme



### CUSTOMER SEGMENTS



Residential solar users (1–10 kW) — 76% of India's 2025 rooftop additions



Small & medium commercial users (10–100 kW)



Solar EPC companies



Agri-solar & irrigation users



158% YoY market growth in 2025



Early adopters in clean-tech ecosystem



TAM: India rooftop solar market valued at \$6.2B (2024), growing to \$19.8B by 2032



### COST STRUCTURE



Hardware manufacturing ₹12,000–16,000/unit BOM



Cloud infrastructure ₹5,000/month AWS



Installation & maintenance costs ₹1,500–2,000/system



Go-to-market (marketing) 8–12% of revenue



Sales & salaries (team) 15–18% of revenue



R&D and prototyping costs ₹2–3 lakh



Marketing & distribution



Salaries and operational expenses



Break-even at ~200 units sold



LTV:CAC target ≥3x



Target gross margin: >40% per unit



### REVENUE STREAMS



Hardware sales ₹20,000–25,000/unit



Installation & maintenance services ₹1,500–2,000/system



B2B bulk contracts (EPC channel)  
Avg. deal size ₹5–20L



India solar tracker market: ₹~1.18B by 2026



Target MRR from SaaS subscriptions ₹10L by FY27



Subscription cloud analytics (SaaS — recurring)  
₹500–1,000/system/month



Warranty packages ₹1,000–2,000/system/year



Govt incentives/subsidies (PM Surya Ghar)



Global tracker market CAGR: 15.56% (2026–2034)



India rooftop additions: 7.1 GW in 2025 (+123% YoY)



### MARKET SIZE & GROWTH



India rooftop solar hit 7.1 GW in 2025, a 123% YoY jump — Mercom India Research, Feb 2026



Cumulative India rooftop solar capacity reached 20.8 GW by end of 2025 — Mercom India / PV Magazine



India rooftop solar market valued at \$6.2B (2024), projected \$19.8B by 2032 at 15.19% CAGR — Maximize Market Research



76% of India's 2025 rooftop additions were residential — Mercom India



158% YoY market growth in installations in 2025 — Maximize Market Research



### SOLAR TRACKER PERFORMANCE

- Trackers deliver 20–30% higher energy yield vs fixed-tilt — Fortune Business Insights / Verified Market Research
- Dual-axis trackers up to 40% gain — multiple tracker market reports
- Indian C&I solar payback period ~3 years — Mordor Intelligence



### TRACKER MARKET OPPORTUNITY

- India solar tracker market projected at ₹~1.18B by 2026 — Fortune Business Insights
- Global solar tracker market CAGR 15.56% (2026–2034)
- India utility-scale tracker adoption rising from 30–35% to 55–60% by FY27 — GameChange Solar CEO, PV Magazine India, Nov 2025



### KEY TAKEAWAY

RayCatcher combines affordable hardware, smart automation, and data-driven insights to maximize solar energy output, improve ROI, and accelerate India's clean energy transition.

